Strategic Use of Information Systems

Recap

1. Definition of MIS
2. Importance of MIS
3. Components of MIS
4. Types of MIS
5. MIS Key Concepts

The Role of IS in Business Strategy

- Achieving competitive advantage
- Enhancing customer experiences
- Enabling innovation

Information Systems (IS) play a crucial role in supporting and shaping business strategies by enabling organizations to respond to market changes, improve internal operations, and deliver value to customers. IS enhances the ability of firms to compete and grow by:

- **Achieving Competitive Advantage**: IS helps firms lower costs, differentiate products/services, and focus on specific market niches. For example, supply chain systems can reduce operational expenses and increase responsiveness.
- **Enhancing Customer Experiences**: IS supports personalized marketing, responsive service, and seamless customer interaction. CRM systems, for example, help companies analyze customer behavior and tailor offerings to meet expectations.
- **Enabling Innovation**: IS is foundational to creating new products, services, and business models. Platforms such as Uber and Airbnb rely entirely on IS to connect users, automate transactions, and scale rapidly.

Information Systems are no longer just tools to automate business processes—they are strategic assets that enable digital transformation and sustainable competitive positioning.

Porter’s Five Forces & IS

- Rivalry among competitors
- Threat of new entrants
- Bargaining power of suppliers
- Bargaining power of customers
- Threat of substitute products
- Role of IS in influencing each force

- **Rivalry Among Existing Competitors:** This force examines the intensity of competition among existing firms in an industry. High rivalry can reduce profit potential.
  - Enterprise Resource Planning (ERP) systems streamline operations and reduce costs.
  - Customer Relationship Management (CRM) systems improve service quality and customer retention.
  - Data analytics help firms understand market trends and adjust strategies quickly.

Example: Amazon uses analytics and AI-driven logistics to outperform competitors in e-commerce.

- **Threat of New Entrants:** This force evaluates how easy or difficult it is for new players to enter an industry.
  - Investment in proprietary platforms and data warehouses can create high entry barriers.
  - Cloud-based services and customer loyalty programs driven by IS discourage market entry.

Example: Netflix uses its recommendation engine and massive content database as barriers to entry.

- **Bargaining Power of Suppliers:** Refers to how much power suppliers have over pricing and availability of materials.
  - IS enables companies to evaluate suppliers, track performance, and source from a global pool.
  - Just-In-Time (JIT) and Supplier Relationship Management (SRM) systems reduce dependence on single suppliers.

Example: Toyota uses IS to manage a diversified and responsive supply chain.

- **Bargaining Power of Buyers:** This force analyzes the power customers have to drive prices down.
  - Personalization engines and loyalty systems improve customer satisfaction and reduce price sensitivity.
  - IS can enable dynamic pricing and tailored promotions to increase perceived value.

Example: Airlines use dynamic pricing systems that adjust fares in real time based on demand.

- **Threat of Substitute Products or Service**: Looks at how easily customers can switch to alternative solutions.
  - Investment in innovation through IS (e.g., mobile apps, AI services) creates unique value propositions.
  - Monitoring market trends through business intelligence tools helps preempt potential substitutes.

Example: Spotify combats substitutes through continuous platform innovation and user analytics.

The Value Chain Model

-a business model that outlines all the activities a company undertakes to deliver a product or service to the customer, from initial idea to final sale and beyond

-using IS to add value

Primary Activities: These are the main activities involved in creating a product or service, including **inbound logistics**, **operations**, **outbound logistics**, **marketing & sales**, and **service**.
Support Activities: These activities support the primary activities and include **procurement**, **technology development**, **human resource management**, and **infrastructure**.

Information Systems contribute value across both **primary** and **support activities** in Porter's Value Chain model:

Primary Activities:

- **Inbound Logistics**: IS improves supply chain visibility, tracks inventory, and automates warehouse operations.
- **Operations**: Enterprise Resource Planning (ERP) systems optimize production schedules, manage resources, and reduce waste.
- **Outbound Logistics**: IS enables real-time delivery tracking, order fulfillment automation, and route optimization.
- **Marketing and Sales**: Digital marketing tools and CRM systems personalize promotions and enhance customer engagement.

- **Service**: IS supports customer support platforms, self-service portals, and feedback systems for better after-sales service.

Support Activities:

- **Firm Infrastructure**: IS streamlines finance, legal, and management functions.
- **Human Resources (HR)**: HRIS systems manage talent acquisition, training, and performance analytics.
- **Technology Development**: IS facilitates R&D through simulation tools and product lifecycle management systems.
- **Procurement**: E-procurement platforms automate vendor selection, bidding, and purchase order processing.

IS enhances efficiency, reduces costs, and creates competitive differentiation in each activity, thus adding strategic value across the organization.

Digital Transformation

-the integration of digital technology into all areas of a business

-to increase efficiency, improve customer experience, drive innovation, and enable agility in response to market changes

-enables companies to adopt new business models

Digital transformation is the integration of digital technology into all areas of a business, fundamentally changing how it operates and delivers value to customers. The primary goals are to increase efficiency, improve customer experience, drive innovation, and enable agility in response to market changes. **Example Technologies**:

- **Cloud Computing**: Facilitates scalable infrastructure, remote access, and cost-effective IT resource management.
- **Artificial Intelligence (AI)**: Enables automation, predictive analytics, and intelligent decision-making.
- **Internet of Things (IoT)**: Connects physical devices to the internet for real-time monitoring, data collection, and process automation.
- **Big Data & Analytics**: Helps organizations analyze large volumes of data to uncover patterns, trends, and insights for better decision-making.

Business Model Innovation: Digital transformation enables companies to adopt new business models—such as platform-based ecosystems, subscription services, and digital marketplaces—that leverage data and technology to deliver superior value. For instance, companies like Netflix and Spotify have shifted from product ownership to service-based streaming models.

Information Systems for Innovation

- Examples: Uber, Airbnb, Zoom
- Creating platform-based business models
- Leveraging data for growth

Information Systems (IS) play a central role in driving business innovation by enabling new ways of creating, delivering, and capturing value. IS fosters innovation by supporting digital platforms, data-driven decision-making, and scalable business models.

Examples of IS-Driven Innovation:

- **Uber**: Uses a digital platform to connect drivers and passengers in real time. Its use of GPS, mobile apps, and data analytics redefined the transportation industry.
- **Airbnb**: Leverages IS to enable peer-to-peer lodging services, disrupting the traditional hospitality industry. Its system manages listings, bookings, payments, and trust through user reviews.
- **Zoom**: Provides a scalable, cloud-based video communication platform. It supports global collaboration, especially during remote work trends, and integrates with enterprise tools.

Creating Platform-Based Business Models:

- IS allows companies to create **platform ecosystems** that bring together

producers and consumers (e.g., app stores, marketplaces, gig platforms).
- These models generate **network effects**, where the value increases as more users join the platform.

Leveraging Data for Growth:

- Businesses use IS to collect and analyze customer behavior, optimize operations, and personalize services.
- Predictive analytics and machine learning help identify trends, segment markets, and improve decision-making.

Activity 2

- choose a local SME
- analyze the business using the Porter’s 5 forces
- propose an IS that could improve their business

1. This is an individual activity.
2. Choose a local SME. Similar company is NOT allowed.
3. Analyze the business using the Porter’s 5 forces. See page 4 for guidance.
4. Create a simple diagram showing the business’ 5 forces.
5. From the business’ 5 forces, propose an IS that could improve their business
6. Present the business’ 5 forces briefly (120 seconds max) in our online class by sharing your screen and presenting your diagram. Include your proposal in the presentation.
7. Submit a 1-page (max) PDF document detailing how your IS proposal will help their business. Also include in the file the business’ 5 forces diagram.
    - Include the class code, name of the class, your name, your section, and date of document creation.
    - Font style: Times New Roman; Font size: 12; single spacing, remove space before and after paragraph.

1\. Deadline of submission is on July 8, 2025, until the end of our online class.